In [2]:
```
import pandas as pd
```

In [7]:
```
df = pd.read_csv('dataset_Facebook.csv',delimiter=';')
```

In [8]:
```
df
```

Out[8]:

	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour	Paid	Lifetime Post Total Reach	Lifetime Post Total Impressions	Lifetime Engaged Users	Co
0	139441	Photo	2	12	4	3	0.0	2752	5091	178	
1	139441	Status	2	12	3	10	0.0	10460	19057	1457	
2	139441	Photo	3	12	3	3	0.0	2413	4373	177	
3	139441	Photo	2	12	2	10	1.0	50128	87991	2211	
4	139441	Photo	2	12	2	3	0.0	7244	13594	671	
5	139441	Status	2	12	1	9	0.0	10472	20849	1191	
6	139441	Photo	3	12	1	3	1.0	11692	19479	481	
7	139441	Photo	3	12	7	9	1.0	13720	24137	537	
8	139441	Status	2	12	7	3	0.0	11844	22538	1530	
9	139441	Photo	3	12	6	10	0.0	4694	8668	280	
10	139441	Status	2	12	5	10	0.0	21744	42334	4258	
11	139441	Photo	2	12	5	10	0.0	3112	5590	208	
12	139441	Photo	2	12	5	10	0.0	2847	5133	193	
13	139441	Photo	2	12	5	3	0.0	2549	4896	249	
14	138414	Photo	2	12	4	5	1.0	22784	39941	887	
15	138414	Status	2	12	3	10	0.0	10060	19680	1264	
16	138414	Photo	3	12	3	3	0.0	1722	2981	163	
17	138414	Photo	1	12	2	12	1.0	53264	111785	1706	
18	138414	Status	3	12	2	3	0.0	3930	7509	130	
19	138414	Photo	3	12	1	11	0.0	1591	2825	121	
20	138414	Photo	2	12	1	3	0.0	2848	5066	200	
21	138414	Photo	1	12	7	10	0.0	1384	2467	15	
22	138414	Link	1	12	7	10	0.0	3454	6853	118	
23	138414	Photo	3	12	7	3	0.0	2723	4888	176	
24	138414	Status	2	12	6	10	0.0	8488	15294	1341	
25	138458	Status	2	12	6	3	0.0	8284	15104	1521	
26	138458	Status	2	12	5	11	0.0	19552	34143	2806	
27	138458	Photo	3	12	5	3	0.0	2478	4306	212	
28	138895	Photo	2	12	5	3	0.0	9560	18264	973	
29	138895	Video	1	12	4	11	1.0	36208	61262	1141	
...	...	...	...	...	...	...	...	...	...	...	
470	91437	Link	1	2	3	13	0.0	9356	14986	448	

471	91330	Photo	3	2	3	3	0.0	7732	13264	1066
472	91223	Photo	1	2	2	13	0.0	5240	8893	857
473	91116	Photo	3	2	2	4	1.0	7132	12060	1004
474	91009	Photo	1	2	1	12	0.0	21928	39641	1512
475	86909	Photo	3	1	6	16	1.0	5754	9238	1179
476	86909	Photo	1	1	6	10	1.0	37088	10966	2728
477	86909	Link	1	1	6	4	0.0	39600	7927	572
478	86909	Photo	3	1	5	13	0.0	5536	8745	1141
479	86909	Photo	3	1	5	4	0.0	6056	10325	1117
480	86909	Photo	2	1	4	11	0.0	11484	20696	1762
481	86491	Link	1	1	4	4	1.0	4938	7910	66
482	86491	Photo	3	1	3	10	0.0	66784	9456	2969
483	86491	Photo	2	1	3	3	0.0	5526	8779	1096
484	86491	Photo	3	1	2	7	0.0	5040	8367	1062
485	86491	Link	1	1	2	2	0.0	5168	8371	66
486	85979	Photo	3	1	1	12	0.0	5034	8030	1020
487	85979	Photo	3	1	1	2	0.0	4908	7491	957
488	85979	Photo	3	1	7	10	0.0	9700	17442	1407
489	85979	Photo	3	1	7	2	0.0	4800	7754	975
490	85979	Photo	3	1	6	11	0.0	5280	8703	951
491	85979	Photo	3	1	6	3	1.0	6184	10228	956
492	85979	Link	1	1	5	11	0.0	45920	5808	753
493	85093	Photo	3	1	1	2	0.0	8412	13960	1179
494	85093	Photo	3	1	7	10	0.0	5400	9218	810
495	85093	Photo	3	1	7	2	0.0	4684	7536	733
496	81370	Photo	2	1	5	8	0.0	3480	6229	537
497	81370	Photo	1	1	5	2	0.0	3778	7216	625
498	81370	Photo	3	1	4	11	0.0	4156	7564	626
499	81370	Photo	2	1	4	4	NaN	4188	7292	564

500 rows × 19 columns

In [10]:
```
df.describe(include='all')
```

Out[10]:

	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour	Paid	Lifetin Total
count	500.000000	500	500.000000	500.000000	500.000000	500.000000	499.000000	50
unique	NaN	4	NaN	NaN	NaN	NaN	NaN	
top	NaN	Photo	NaN	NaN	NaN	NaN	NaN	
freq	NaN	426	NaN	NaN	NaN	NaN	NaN	
mean	123194.176000	NaN	1.880000	7.038000	4.150000	7.840000	0.278557	1390
std	16272.813214	NaN	0.852675	3.307936	2.030701	4.368589	0.448739	2274
min	81370.000000	NaN	1.000000	1.000000	1.000000	1.000000	0.000000	23
25%	112676.000000	NaN	1.000000	4.000000	2.000000	3.000000	0.000000	331
50%	129600.000000	NaN	2.000000	7.000000	4.000000	9.000000	0.000000	528
75%	136393.000000	NaN	3.000000	10.000000	6.000000	11.000000	1.000000	1316
max	139441.000000	NaN	3.000000	12.000000	7.000000	23.000000	1.000000	18048

In [11]:
```
df_subset_1 = df[['like','share']]
df_subset_1
```

Out[11]:

	like	share
0	79.0	17.0
1	130.0	29.0
2	66.0	14.0
3	1572.0	147.0
4	325.0	49.0
5	152.0	33.0
6	249.0	27.0
7	325.0	14.0
8	161.0	31.0
9	113.0	26.0
10	233.0	19.0
11	88.0	18.0
12	90.0	14.0
13	137.0	10.0
14	577.0	20.0
15	86.0	18.0
16	40.0	12.0
17	678.0	20.0
18	54.0	17.0
19	34.0	8.0
20	66.0	12.0
21	0.0	0.0
22	16.0	2.0
23	72.0	24.0
24	99.0	19.0
25	88.0	18.0
26	412.0	72.0
27	100.0	17.0
28	523.0	63.0
29	143.0	13.0
...	...	...
470	114.0	13.0
471	160.0	57.0
472	46.0	15.0
473	136.0	42.0
474	73.0	13.0
475	65.0	19.0
476	579.0	47.0

477	101.0	5.0
478	74.0	31.0
479	84.0	36.0
480	360.0	99.0
481	5.0	2.0
482	187.0	36.0
483	69.0	26.0
484	82.0	24.0
485	12.0	2.0
486	56.0	25.0
487	44.0	21.0
488	277.0	80.0
489	74.0	28.0
490	79.0	30.0
491	105.0	46.0
492	128.0	9.0
493	185.0	55.0
494	125.0	41.0
495	53.0	26.0
496	53.0	22.0
497	93.0	18.0
498	91.0	38.0
499	91.0	28.0

500 rows × 2 columns

In [12]:
```
df_subset_2 = df[['comment','Type']]
df_subset_2
```

Out[12]:

	comment	Type
0	4	Photo
1	5	Status
2	0	Photo
3	58	Photo
4	19	Photo
5	1	Status
6	3	Photo
7	0	Photo
8	0	Status
9	3	Photo
10	0	Status
11	0	Photo
12	0	Photo
13	5	Photo
14	2	Photo
15	4	Status
16	2	Photo
17	15	Photo
18	4	Status
19	0	Photo
20	3	Photo
21	0	Photo
22	0	Link
23	0	Photo
24	3	Status
25	0	Status
26	10	Status
27	0	Photo
28	36	Photo
29	18	Video
...	...	...
470	4	Link
471	7	Photo
472	0	Photo
473	4	Photo
474	0	Photo
475	0	Photo
476	7	Photo

477	1	Link
478	1	Photo
479	7	Photo
480	56	Photo
481	0	Link
482	3	Photo
483	2	Photo
484	2	Photo
485	0	Link
486	2	Photo
487	1	Photo
488	21	Photo
489	1	Photo
490	1	Photo
491	1	Photo
492	0	Link
493	17	Photo
494	10	Photo
495	5	Photo
496	0	Photo
497	4	Photo
498	7	Photo
499	0	Photo

500 rows × 2 columns

In [13]:
```
merged_data  = pd.merge(df_subset_2, df_subset_1, left_on='comment', ri
ght_on= 'like')
merged_data
```

Out[13]:

	comment	Type	like	share
0	4	Photo	4.0	2.0
1	4	Photo	4.0	1.0
2	4	Photo	4.0	0.0
3	4	Photo	4.0	1.0
4	4	Status	4.0	2.0
5	4	Status	4.0	1.0
6	4	Status	4.0	0.0
7	4	Status	4.0	1.0
8	4	Status	4.0	2.0
9	4	Status	4.0	1.0
10	4	Status	4.0	0.0
11	4	Status	4.0	1.0
12	4	Photo	4.0	2.0
13	4	Photo	4.0	1.0
14	4	Photo	4.0	0.0
15	4	Photo	4.0	1.0
16	4	Link	4.0	2.0
17	4	Link	4.0	1.0
18	4	Link	4.0	0.0
19	4	Link	4.0	1.0
20	4	Status	4.0	2.0
21	4	Status	4.0	1.0
22	4	Status	4.0	0.0
23	4	Status	4.0	1.0
24	4	Photo	4.0	2.0
25	4	Photo	4.0	1.0
26	4	Photo	4.0	0.0
27	4	Photo	4.0	1.0
28	4	Photo	4.0	2.0
29	4	Photo	4.0	1.0
...	...	...	...	...
1437	22	Photo	22.0	6.0
1438	144	Photo	144.0	10.0
1439	144	Photo	144.0	29.0
1440	38	Status	38.0	11.0
1441	29	Photo	29.0	10.0
1442	29	Photo	29.0	7.0
1443	41	Status	41.0	9.0

1444	41	Status	41.0	9.0
1445	64	Photo	64.0	19.0
1446	64	Photo	64.0	7.0
1447	64	Photo	64.0	20.0
1448	64	Photo	64.0	22.0
1449	23	Video	23.0	2.0
1450	25	Photo	25.0	5.0
1451	25	Photo	25.0	6.0
1452	25	Photo	25.0	5.0
1453	25	Photo	25.0	6.0
1454	25	Photo	25.0	5.0
1455	25	Photo	25.0	6.0
1456	37	Photo	37.0	3.0
1457	51	Photo	51.0	11.0
1458	51	Photo	51.0	6.0
1459	51	Photo	51.0	6.0
1460	146	Photo	146.0	9.0
1461	146	Photo	146.0	15.0
1462	56	Photo	56.0	17.0
1463	56	Photo	56.0	8.0
1464	56	Photo	56.0	12.0
1465	56	Photo	56.0	9.0
1466	56	Photo	56.0	25.0

1467 rows × 4 columns

In [14]:
```
# Define a dictionary containing employee data
data1 = {
'key': ['K0', 'K1', 'K2', 'K3'],
'Name':['Jai', 'Princi', 'Gaurav', 'Anuj'],
'Age':[27, 24, 22, 32],}

data2 = {
'key': ['K0', 'K1', 'K2', 'K3'],
'Address':['Nagpur', 'Kanpur', 'Allahabad', 'Kannuaj'],
'Qualification':['Btech', 'B.A', 'Bcom', 'B.hons']}

data1 = pd.DataFrame(data1)

data2 = pd.DataFrame(data2)

res = pd.merge(data1, data2, on='key')
res
```

Out[14]:

	key	Name	Age	Address	Qualification
0	K0	Jai	27	Nagpur	Btech
1	K1	Princi	24	Kanpur	B.A
2	K2	Gaurav	22	Allahabad	Bcom
3	K3	Anuj	32	Kannuaj	B.hons

In [15]:
```
merged_data.sort_values(by=['like'],ascending=False)
```

Out[15]:

	comment	Type	like	share
1461	146	Photo	146.0	15.0
1460	146	Photo	146.0	9.0
1438	144	Photo	144.0	10.0
1439	144	Photo	144.0	29.0
1448	64	Photo	64.0	22.0
1445	64	Photo	64.0	19.0
1446	64	Photo	64.0	7.0
1447	64	Photo	64.0	20.0
1423	60	Status	60.0	7.0
727	58	Photo	58.0	13.0
726	58	Photo	58.0	17.0
1466	56	Photo	56.0	25.0
1463	56	Photo	56.0	8.0
1464	56	Photo	56.0	12.0
1465	56	Photo	56.0	9.0
1462	56	Photo	56.0	17.0
1459	51	Photo	51.0	6.0
1457	51	Photo	51.0	11.0
1458	51	Photo	51.0	6.0
1425	47	Photo	47.0	8.0
1426	47	Photo	47.0	4.0
1427	47	Photo	47.0	21.0
1424	47	Photo	47.0	11.0
1422	42	Photo	42.0	10.0
1444	41	Status	41.0	9.0
1443	41	Status	41.0	9.0
1440	38	Status	38.0	11.0
1456	37	Photo	37.0	3.0
1122	36	Photo	36.0	11.0
1126	36	Status	36.0	5.0
...	...	...	...	...
356	0	Photo	0.0	0.0
357	0	Photo	0.0	0.0
358	0	Photo	0.0	0.0
359	0	Photo	0.0	0.0
360	0	Photo	0.0	0.0
361	0	Photo	0.0	0.0
362	0	Photo	0.0	0.0

363	0	Photo	0.0	0.0
364	0	Photo	0.0	0.0
365	0	Photo	0.0	0.0
366	0	Photo	0.0	0.0
367	0	Photo	0.0	0.0
368	0	Photo	0.0	0.0
369	0	Photo	0.0	0.0
370	0	Photo	0.0	0.0
371	0	Photo	0.0	0.0
372	0	Photo	0.0	0.0
373	0	Photo	0.0	0.0
556	0	Photo	0.0	0.0
557	0	Photo	0.0	0.0
558	0	Photo	0.0	0.0
559	0	Photo	0.0	0.0
560	0	Photo	0.0	0.0
561	0	Photo	0.0	0.0
562	0	Photo	0.0	0.0
563	0	Photo	0.0	0.0
564	0	Photo	0.0	0.0
565	0	Photo	0.0	0.0
566	0	Photo	0.0	0.0
380	0	Photo	0.0	0.0

1467 rows × 4 columns

In [16]:
```
merged_data.transpose()
#method 1
```

Out[16]:

	0	1	2	3	4	5	6	7	8	9	...	1457	145
comment	4	4	4	4	4	4	4	4	4	4	...	51	5
Type	Photo	Photo	Photo	Photo	Status	Status	Status	Status	Status	Status	...	Photo	Pho
like	4	4	4	4	4	4	4	4	4	4	...	51	5
share	2	1	0	1	2	1	0	1	2	1	...	11	

4 rows × 1467 columns

In [17]:
```
merged_data.T
#method 2
```

Out[17]:

	0	1	2	3	4	5	6	7	8	9	...	1457	145
comment	4	4	4	4	4	4	4	4	4	4	...	51	5
Type	Photo	Photo	Photo	Photo	Status	Status	Status	Status	Status	Status	...	Photo	Pho
like	4	4	4	4	4	4	4	4	4	4	...	51	5
share	2	1	0	1	2	1	0	1	2	1	...	11	

4 rows × 1467 columns

In [18]:

```
df
```

Out[18]:

	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour	Paid	Lifetime Post Total Reach	Lifetime Post Total Impressions	Lifetime Engaged Users	Co
0	139441	Photo	2	12	4	3	0.0	2752	5091	178	
1	139441	Status	2	12	3	10	0.0	10460	19057	1457	
2	139441	Photo	3	12	3	3	0.0	2413	4373	177	
3	139441	Photo	2	12	2	10	1.0	50128	87991	2211	
4	139441	Photo	2	12	2	3	0.0	7244	13594	671	
5	139441	Status	2	12	1	9	0.0	10472	20849	1191	
6	139441	Photo	3	12	1	3	1.0	11692	19479	481	
7	139441	Photo	3	12	7	9	1.0	13720	24137	537	
8	139441	Status	2	12	7	3	0.0	11844	22538	1530	
9	139441	Photo	3	12	6	10	0.0	4694	8668	280	
10	139441	Status	2	12	5	10	0.0	21744	42334	4258	
11	139441	Photo	2	12	5	10	0.0	3112	5590	208	
12	139441	Photo	2	12	5	10	0.0	2847	5133	193	
13	139441	Photo	2	12	5	3	0.0	2549	4896	249	
14	138414	Photo	2	12	4	5	1.0	22784	39941	887	
15	138414	Status	2	12	3	10	0.0	10060	19680	1264	
16	138414	Photo	3	12	3	3	0.0	1722	2981	163	
17	138414	Photo	1	12	2	12	1.0	53264	111785	1706	
18	138414	Status	3	12	2	3	0.0	3930	7509	130	
19	138414	Photo	3	12	1	11	0.0	1591	2825	121	
20	138414	Photo	2	12	1	3	0.0	2848	5066	200	
21	138414	Photo	1	12	7	10	0.0	1384	2467	15	
22	138414	Link	1	12	7	10	0.0	3454	6853	118	
23	138414	Photo	3	12	7	3	0.0	2723	4888	176	
24	138414	Status	2	12	6	10	0.0	8488	15294	1341	
25	138458	Status	2	12	6	3	0.0	8284	15104	1521	
26	138458	Status	2	12	5	11	0.0	19552	34143	2806	
27	138458	Photo	3	12	5	3	0.0	2478	4306	212	
28	138895	Photo	2	12	5	3	0.0	9560	18264	973	
29	138895	Video	1	12	4	11	1.0	36208	61262	1141	
...	...	...	...	...	...	...	...	...	...	...	
470	91437	Link	1	2	3	13	0.0	9356	14986	448	

471	91330	Photo	3	2	3	3	0.0	7732	13264	1066
472	91223	Photo	1	2	2	13	0.0	5240	8893	857
473	91116	Photo	3	2	2	4	1.0	7132	12060	1004
474	91009	Photo	1	2	1	12	0.0	21928	39641	1512
475	86909	Photo	3	1	6	16	1.0	5754	9238	1179
476	86909	Photo	1	1	6	10	1.0	37088	10966	2728
477	86909	Link	1	1	6	4	0.0	39600	7927	572
478	86909	Photo	3	1	5	13	0.0	5536	8745	1141
479	86909	Photo	3	1	5	4	0.0	6056	10325	1117
480	86909	Photo	2	1	4	11	0.0	11484	20696	1762
481	86491	Link	1	1	4	4	1.0	4938	7910	66
482	86491	Photo	3	1	3	10	0.0	66784	9456	2969
483	86491	Photo	2	1	3	3	0.0	5526	8779	1096
484	86491	Photo	3	1	2	7	0.0	5040	8367	1062
485	86491	Link	1	1	2	2	0.0	5168	8371	66
486	85979	Photo	3	1	1	12	0.0	5034	8030	1020
487	85979	Photo	3	1	1	2	0.0	4908	7491	957
488	85979	Photo	3	1	7	10	0.0	9700	17442	1407
489	85979	Photo	3	1	7	2	0.0	4800	7754	975
490	85979	Photo	3	1	6	11	0.0	5280	8703	951
491	85979	Photo	3	1	6	3	1.0	6184	10228	956
492	85979	Link	1	1	5	11	0.0	45920	5808	753
493	85093	Photo	3	1	1	2	0.0	8412	13960	1179
494	85093	Photo	3	1	7	10	0.0	5400	9218	810
495	85093	Photo	3	1	7	2	0.0	4684	7536	733
496	81370	Photo	2	1	5	8	0.0	3480	6229	537
497	81370	Photo	1	1	5	2	0.0	3778	7216	625
498	81370	Photo	3	1	4	11	0.0	4156	7564	626
499	81370	Photo	2	1	4	4	NaN	4188	7292	564

500 rows × 19 columns

In [19]:
```
df.Type.unique()
```

Out[19]:
```
array(['Photo', 'Status', 'Link', 'Video'], dtype=object)
```

In [20]:
```
# Reshape
pd.melt(df, id_vars =['Type'], value_vars =['comment'])
```

Out[20]:

	Type	variable	value
0	Photo	comment	4
1	Status	comment	5
2	Photo	comment	0
3	Photo	comment	58
4	Photo	comment	19
5	Status	comment	1
6	Photo	comment	3
7	Photo	comment	0
8	Status	comment	0
9	Photo	comment	3
10	Status	comment	0
11	Photo	comment	0
12	Photo	comment	0
13	Photo	comment	5
14	Photo	comment	2
15	Status	comment	4
16	Photo	comment	2
17	Photo	comment	15
18	Status	comment	4
19	Photo	comment	0
20	Photo	comment	3
21	Photo	comment	0
22	Link	comment	0
23	Photo	comment	0
24	Status	comment	3
25	Status	comment	0
26	Status	comment	10
27	Photo	comment	0
28	Photo	comment	36
29	Video	comment	18
...	...	...	...
470	Link	comment	4
471	Photo	comment	7
472	Photo	comment	0
473	Photo	comment	4
474	Photo	comment	0
475	Photo	comment	0
476	Photo	comment	7

477	Link	comment	1
478	Photo	comment	1
479	Photo	comment	7
480	Photo	comment	56
481	Link	comment	0
482	Photo	comment	3
483	Photo	comment	2
484	Photo	comment	2
485	Link	comment	0
486	Photo	comment	2
487	Photo	comment	1
488	Photo	comment	21
489	Photo	comment	1
490	Photo	comment	1
491	Photo	comment	1
492	Link	comment	0
493	Photo	comment	17
494	Photo	comment	10
495	Photo	comment	5
496	Photo	comment	0
497	Photo	comment	4
498	Photo	comment	7
499	Photo	comment	0

500 rows × 3 columns

In [21]:
```
# Reshape
df_temp = pd.DataFrame({'foo': ['one', 'one', 'one', 'two', 'two', 'tw
o'],
                    'bar': ['A', 'B', 'C', 'A', 'B', 'C'],
                    'baz': [1, 2, 3, 4, 5, 6],
                    'zoo': ['x', 'y', 'z', 'q', 'w', 't']})
df_temp
```

Out[21]:

	foo	bar	baz	zoo
0	one	A	1	x
1	one	B	2	y
2	one	C	3	z
3	two	A	4	q
4	two	B	5	w
5	two	C	6	t

In [22]:
```
df_temp.pivot(index='foo', columns='bar', values='baz')
```

Out[22]:

bar	A	B	C
foo			
one	1	2	3
two	4	5	6

In [ ]: